

Chapter 2 — Finite Markov Decision Processes

Question 1 — Discounted Return

An agent receives the following sequence of rewards:

$$R_{t+1} = 4, \quad R_{t+2} = 3, \quad R_{t+3} = 6, \quad R_{t+4} = 2$$

The discount rate is

$$\gamma = 0.8.$$

Calculate the discounted return G_t .

Question 2 — Effect of the Discount Factor

An agent receives a reward of 10 after 1 step, 8 after 2 steps, and 6 after 3 steps.

Calculate the return for each of the following discount rates:

1. $\gamma = 0$
2. $\gamma = 0.5$
3. $\gamma = 0.9$

Use:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3}.$$

Which discount rate makes the agent more sensitive to future rewards?

Question 3 — State-Value Function

Consider an MDP with two states, A and B . Under policy π :

- From state A , the agent receives a reward of 4 and moves to B with probability 1.
- From state B , the agent receives a reward of 2 and remains in B with probability 1.
- The discount factor is $\gamma = 0.5$.

Using the Bellman equation,

$$v^\pi(s) = E_\pi[R_{t+1} + \gamma v^\pi(S_{t+1}) | S_t = s],$$

calculate:

$$v^\pi(A)$$

and

$$v^\pi(B).$$

Question 4 — Bellman Equation with Probabilistic Transitions

An agent is in state S_1 and follows a fixed policy. The environment behaves as follows:

Next State	Probability	Reward
S_1	0.4	2
S_2	0.6	5

Suppose:

$$v^\pi(S_1) = 10, \quad v^\pi(S_2) = 20, \quad \gamma = 0.9.$$

Calculate $v^\pi(S_1)$ using the Bellman expectation equation:

$$v^\pi(s) = \sum_{s', r} p(s', r | s, a) [r + \gamma v^\pi(s')].$$

Question 5 — Action-Value Function

An agent is currently in state S and takes action a .

The possible outcomes are:

- With probability 0.7, the agent receives reward 5 and moves to state S_1 .
- With probability 0.3, the agent receives reward -2 and moves to state S_2 .

Suppose:

$$v^\pi(S_1) = 8, \quad v^\pi(S_2) = 4,$$

and

$$\gamma = 0.8.$$

Calculate the action-value:

$$q^\pi(S, a) = E[R_{t+1} + \gamma v^\pi(S_{t+1}) | S_t = S, A_t = a].$$

Question 6 — Bellman Optimality Equation

An MDP has two possible actions in state S : a_1 and a_2 .

For action a_1 :

- reward = 4
- next state S_1 with probability 1

For action a_2 :

- reward = 7
- next state S_2 with probability 1

Suppose:

$$v^*(S_1) = 10, \quad v^*(S_2) = 5,$$

and

$$\gamma = 0.9.$$

Using the Bellman optimality equation,

$$v^*(S) = \max_a E[R_{t+1} + \gamma v^*(S_{t+1}) | S_t = S, A_t = a],$$

calculate $v^*(S)$ and determine the optimal action.

Question 7 — Bellman Optimality with Probabilities

Consider two actions available in state S .

For action a_1 :

$$P(S_1|S, a_1) = 0.6, \quad P(S_2|S, a_1) = 0.4$$

and the rewards are 4 and 2, respectively.

For action a_2 :

$$P(S_1|S, a_2) = 0.3, \quad P(S_2|S, a_2) = 0.7$$

and the rewards are 6 and 1, respectively.

Given:

$$v^*(S_1) = 12, \quad v^*(S_2) = 8, \quad \gamma = 0.8.$$

Calculate the expected value of each action and determine the optimal action.

Question 8 — Recycling Robot

A recycling robot has two states:

$$S = \{High, Low\}.$$

Suppose:

$$\begin{aligned} \alpha &= 0.8, & \beta &= 0.6, \\ r_{search} &= 2, & r_{wait} &= 1, \end{aligned}$$

and

$$\gamma = 0.9.$$

The Bellman optimality equations are:

$$v^*(High) = \max \{2 + 0.9[0.8v^*(High) + 0.2v^*(Low)], \quad 1 + 0.9v^*(High)\}$$

and

$$v^*(Low) = \max \{0.6(2) - 3(0.4) + 0.9[0.4v^*(High) + 0.6v^*(Low)], \quad 1 + 0.9v^*(Low), \quad 0.9v^*(High)\}$$

Determine the optimal action in each state and calculate $v^*(High)$ and $v^*(Low)$.

Question 9 — Comparing Two Policies

An MDP has two policies, π_1 and π_2 . Their state-value functions are:

State	$v^{\pi_1}(s)$	$v^{\pi_2}(s)$
S_1	12	15
S_2	8	8
S_3	20	18

Using the definition of policy ordering,

$$\pi_1 \geq \pi_2 \quad \text{iff} \quad v^{\pi_1}(s) \geq v^{\pi_2}(s)$$

for **every state**, determine whether $\pi_1 \geq \pi_2$, $\pi_2 \geq \pi_1$, or neither.

Question 10 — Optimal Action-Value Function

An MDP has the following action values for state S :

$$q^*(S, a_1) = 12.5$$

$$q^*(S, a_2) = 15.2$$

$$q^*(S, a_3) = 14.7.$$

Using

$$v^*(S) = \max_a q^*(S, a),$$

calculate $v^*(S)$ and identify the optimal action.

Question 11 — Markov Property

Consider an environment where the next state depends on the **current position and current velocity**, but not on how the agent reached that position.

At time t :

$$S_t = (x_t, v_t) = (5, 2).$$

Two different histories lead to exactly the same state:

- History 1: $x = 1 \rightarrow 3 \rightarrow 5$
- History 2: $x = 4 \rightarrow 3 \rightarrow 5$

The current velocity in both cases is 2.

If

$$P(S_{t+1} = S' | S_t = (5, 2), A_t = a) = 0.7$$

for both histories, determine whether the state representation satisfies the Markov property. **Explain using the Markov definition.**

Question 12 —

A recycling robot has two states:

$$S = \{High, Low\}.$$

Suppose:

$$\alpha = 0.8, \quad \beta = 0.6,$$

$$r_{search} = 2, \quad r_{wait} = 1,$$

and

$$\gamma = 0.9.$$

The Bellman optimality equations are:

$$v^*(High) = \max \{2 + 0.9[0.8v^*(High) + 0.2v^*(Low)], \quad 1 + 0.9v^*(High)\}$$

and

$$v^*(Low) = \max \{0.6(2) - 3(0.4) + 0.9[0.4v^*(High) + 0.6v^*(Low)], \quad 1 + 0.9v^*(Low), \quad 0.9v^*(High)\}$$

Determine the optimal action in each state and calculate $v^*(High)$ and $v^*(Low)$.